

ABEER'S TECH SOLUTIONS

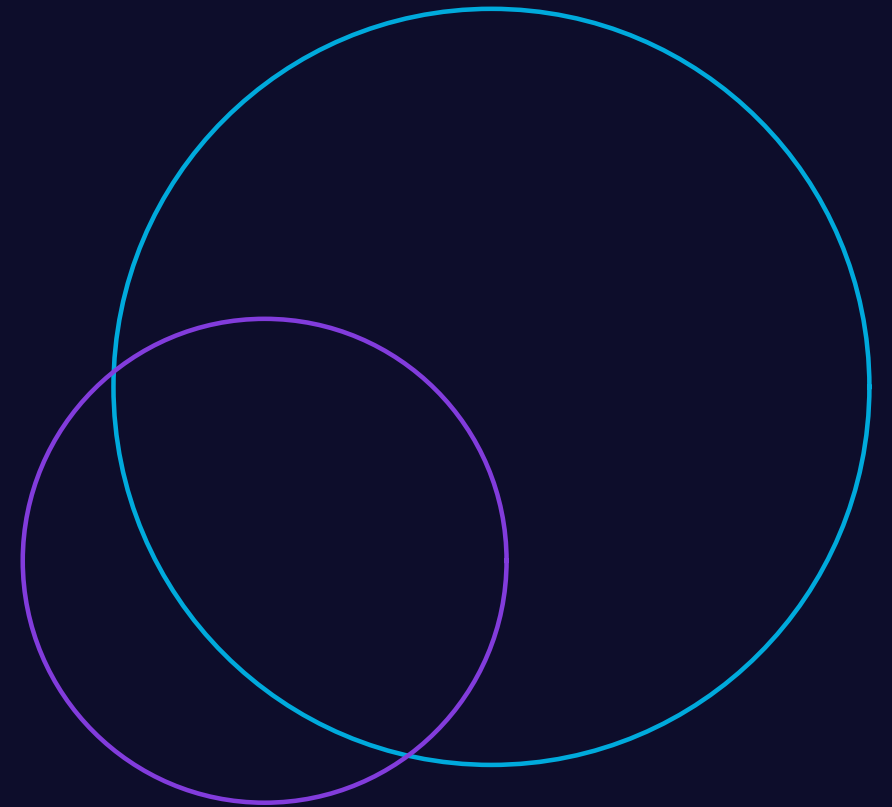
# Goethe Booking Bot

Automated Exam Registration System for Goethe-Institut Pakistan

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**Presented to: Hamza**

Presented by: Abeer | June 2026



# Agenda

What we will cover today

## 01 | The Problem

Why manual booking fails

## 02 | The Solution

What the bot does

## 03 | System Architecture

How everything connects

## 04 | Bot Workflow

6 phases from start to confirmation

## 05 | Multi-Student Execution

3 students running simultaneously

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## 07 | Error Handling

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## 08 | Summary & Next Steps

How to proceed

# The Problem

Why manual booking fails every single time



## Limited Time Window

Registration opens 1 day per month for only 30 minutes. Miss it and you wait another month for the next slot.



## Website Overload

Hundreds of students login simultaneously. The server crashes with 502/503 errors, wasting your precious seconds.



## Human Slowness

A human takes 30-60 seconds to notice the button, click, login, and fill the form. By then all slots are gone.

**The result: Most students fail to book every month - even with perfect internet and multiple devices.**

# The Solution

Goethe Booking Bot - your 24/7 automated booker

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## Fully Automated

One click to start. Bot runs 24/7 on the cloud. No manual intervention needed on exam day.

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## 10x Faster

Reacts in milliseconds when booking opens. Fills entire registration form in under 5 seconds.

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## Multi-Student

Handles 3+ students simultaneously with separate Chrome browsers running in parallel.

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## Error-Resilient

Auto-retries on crashes, timeouts, and server errors. Never gives up until successful.

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## Cloud-Powered

Deployed on Railway cloud servers. Your computer, phone, or tablet can stay completely OFF.

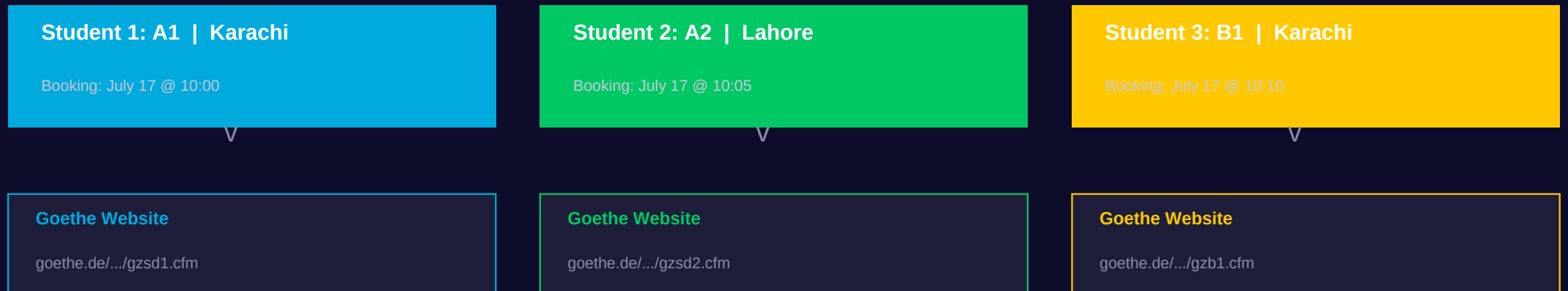
**You click Start. The bot does everything else. From monitoring to booking confirmation.**

# System Architecture

From your browser to the Goethe website



3 Chrome browsers - one per student:



Each Chrome visits its own exam page | Each runs independently | They never interfere

# Bot Workflow

6 phases - from idle monitoring to confirmed booking



PHASE 1

**Waiting**

Polls every 45s



PHASE 2

**Burst Mode**

Polls every 2-3s



PHASE 3

**Auto-Login**

My Goethe.de CAS



PHASE 4

**Form Fill**

6 fields in 2s



PHASE 5

**Submit**

Confirm booking



PHASE 6

**Done**

Screen + notify

## Phase 1: Waiting

Opens exam page. Checks every 45 seconds for the "Book Now" button. Can run silently for days or weeks.

## Phase 2: Burst Mode

30 seconds before booking time, polling speeds up to every 2-3 seconds. Maximum alertness.

## Phase 3: Auto-Login

"Book Now" clicked instantly. Redirected to My Goethe.de CAS login. Credentials auto-filled.

## Phase 4: Form Filling

All 6 registration fields completed in under 2 seconds. No typos, no corrections.

## Phase 5: Submit

Form submitted. Confirmation page screenshot captured as proof of booking.

## Phase 6: Notification

Telegram message sent with booking status and reference number.

# Phase 1: Waiting Mode

24/7 monitoring - runs for days until booking opens

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## Polling Loop

Each Chrome visits its Goethe exam page every 45 seconds and checks for the "Book Now" button. Minimal CPU usage.

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## Zero Manual Input

Bot runs on Railway cloud servers. Your computer, laptop, and phone can be completely turned OFF.

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## Start Days in Advance

Bot can be started 1-2 weeks before the booking opens. It patiently waits and never misses the moment.

```
[INFO] Bot started with 3 student(s)

[INFO] STARTING STUDENT: Abeer Meer | A1 | Karachi

[INFO] Monitoring URL: goethe.de/.../gzsd1.cfm

[INFO] Booking time: 2026-07-17T10:00:00

[INFO] STEP 1: Waiting for Book Now button
```

**Example: Start bot on July 15. It polls every 45s for 2 days until "Book Now" appears on July 17.**

## Phase 2: Burst Mode

The 30-second window where the race is won or lost

### Human

Notice button: 3-10 seconds

Reach for mouse: 1-2 seconds

Click button: 0.5 seconds

**5-12 seconds total**

### Bot

Detect button: 0.3 seconds

Click button: 0.1 seconds

No hesitation, no delay

**0.4 seconds total**

>>

#### 502/503 Errors? No Problem

If server crashes during burst, bot auto-refreshes within 5 seconds and keeps trying.

>>

#### 12-30x Faster Than Human

In the first 10 seconds of booking, bot has already clicked, logged in, and is filling the form.



# Phase 3: Auto-Login

My Goethe.de CAS login - fully automated

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## Step 1: Detect Login Page

After clicking "Book Now", bot is redirected to My Goethe.de CAS login page. Waits for form to fully load.

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## Step 2: Auto-Fill Credentials

Email and password are typed automatically with realistic keystroke timing. Human: 5-8s | Bot: 0.5s

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## Step 3: Submit & Proceed

Login button clicked. Bot waits for registration form. If login fails, retries automatically.

**Credentials stored in config.csv - never committed to GitHub. Your password stays safe.**

# Phase 4: Form Filling

6 fields completed in under 2 seconds - no errors, no typos

## Registration Form Fields

1. Name - Single field
2. Date of Birth - Auto-formatted
3. Place of Birth - Full city name
4. Address - Full street address
5. Phone - With country code
6. Level - A1, A2, or B1 dropdown

## Speed Comparison

Human: 20-40 seconds

Bot: 1.5-2 seconds



No typos | No backspace | No corrections

Date format auto-validated

Phone format auto-corrected

If Goethe changes the form layout, bot saves a screenshot - share it, selectors updated within minutes.

# Phase 5: Submit & Confirm

Booking confirmation captured and saved

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**Form Submission**  
  
Submit button clicked. Bot waits for confirmation page.  
Always verifies the result.

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**Screenshot Captured**  
  
Confirmation page is automatically screenshotted as  
proof of successful booking.

>>

**Notification Sent**  
  
Telegram message sent with booking status and  
reference number.

## Possible Outcomes:

CONFIRMED

Reference received. Booking 100% successful.

SUBMITTED

Form sent. Reference via email later.

FAILED

Error logged. No silent failures.

STOPPED

User clicked Stop. Exits gracefully.

# Multi-Student Execution

3 students, 3 Chrome browsers - all running simultaneously

## STUDENT 1

### A1 | Karachi

gzsd1.cfm

Booking: July 17 @ 10:00

Status: Waiting...

## STUDENT 2

### A2 | Lahore

gzsd2.cfm

Booking: July 17 @ 10:05

Status: Waiting...

## STUDENT 3

### B1 | Karachi

gzb1.cfm

Booking: July 17 @ 10:10

Status: Waiting...

- > Each student gets a dedicated Chrome instance with separate profiles, cookies, and sessions.
- > They never interfere with each other. Each runs fully independently in its own browser.
- > Add more students easily - just update config.csv. Works with Karachi, Lahore, or Islamabad.

# Speed Comparison

Bot vs Human - every millisecond counts in a 30-minute window

| Action                    | Human     | Bot              | Speedup |
|---------------------------|-----------|------------------|---------|
| Detect "Book Now" button  | 3-10 sec  | 0.3 sec          | 10-33x  |
| Click the button          | 1-2 sec   | 0.1 sec          | 10-20x  |
| Navigate to login page    | 3-5 sec   | 1-2 sec          | 2-3x    |
| Type email + password     | 5-8 sec   | 0.5 sec          | 10-16x  |
| Fill registration form    | 20-40 sec | 1.5 sec          | 13-27x  |
| Submit & get confirmation | 2-5 sec   | 0.5 sec          | 4-10x   |
| TOTAL TIME                | 35-70 sec | 4-5 sec          |         |
| Human: 35-70 seconds      |           | Bot: 4-5 seconds |         |
| Human 93%                 |           |                  | Bot 7%  |

The bot is 8-15x faster. In a 30-minute booking window, those seconds are everything.

# Error Handling & Resilience

Built to survive server crashes, network failures, and Chrome issues

## 502 / 503 Server Error

Bot auto-refreshes after 3-5 seconds and re-checks. Keeps trying until button appears.

## Chrome Crash (OOM)

Auto-restarts Chrome (up to 3 attempts, 5s apart). Other students are unaffected.

## Login Failure

Re-enters credentials and retries. Logs error and moves on if persistent.

## Form Submit Timeout

Restarts entire flow from "Book Now" detection if page times out.

## Network Disconnect

Detects connection loss. Automatically refreshes when connection restores.

**The bot never gives up. It keeps trying until successful, stopped, or the booking window expires.**

# Deployment & 24/7 Operation

Cloud-hosted - your devices stay free



## Cloud-Hosted 24/7

Deployed on Railway servers. Runs 24/7 with zero downtime. Your computer stays OFF.



## Real-Time Dashboard

Netlify web UI shows live student status, logs, and results. Accessible from any browser.



## One-Click Start

Open dashboard URL and click Start. Bot takes over. Nothing to install or configure.



## Access Anywhere

Works on phone, tablet, laptop - any device with a browser. No app needed.



## Auto-Updates

Code improvements via GitHub push. Railway auto-rebuilds and redeploys in minutes.



## Secure by Design

Credentials never stored in GitHub. Config uploaded separately via secure API.

# Web Dashboard Preview

What you see when you open the URL

Goethe Booking Bot

Running

Abeer Meer

A1 | Karachi

Waiting...

Abeer Meer

A2 | Lahore

Waiting...

Abeer Meer

B1 | Karachi

Waiting...

[INFO] Bot started with 3 student(s)

[INFO] STARTING STUDENT: Abeer Meer | A1 | Karachi

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[INFO] STEP 1: Waiting for Book Now button



# Why Choose This Solution?

Built by someone who understands the struggle

## > Purpose-Built for Goethe PK

Not a generic bot. Built specifically for Goethe-Institut Pakistan. Handles Karachi, Lahore, and Islamabad with ease.

## > Truly Hands-Free

Once started, needs zero manual input. Runs 24/7 on cloud. Monitor from your phone anytime.

## > Multi-Student Support

3+ students at once. Each runs independently. Add more students anytime by updating config.

## > Proven Speed Advantage

Bot is 8-15x faster than any human. Reacts in milliseconds. Survives server crashes with auto-retry.

## > Lifetime Maintenance

If Goethe changes their website, I update the bot within hours. You never worry about it breaking.

## > Direct Support

You get direct access to me (Abeer). No support tickets, no chatbots, no automated replies.

# Summary & Next Steps

Ready to never miss a Goethe booking again

## Problem

Goethe booking opens 30 min once a month. Server crashes. Humans are too slow.

## Solution

Automated bot monitors 24/7, reacts in milliseconds, fills forms in 2 seconds.

## Architecture

3 Chrome browsers on Railway cloud, controlled via Netlify web dashboard.

## Speed

8-15x faster than human. Sub-second reaction. 4-5 second total flow time.

## Resilience

Survives 502/503 errors, Chrome crashes, network drops. Auto-retries everything.

## Deployment

Go live in 5 minutes. Start bot, relax, check results later.

## Next Steps:

Confirm scope > Customize config > Deploy > Test > Ready for next booking date!

Thank you, Hamza